Are we underestimating Paleoindian populations?

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Introduction: Dates-as-data and landscape taphonomy

Accurately reconstructing past human population dynamics is critical for explaining major patterns in the human past. Demand for demographic proxies has driven hopeful interest in the "dates-as-data" approach (Rick 1987), which models past demography by assuming a relationship between population size, the production of dateable material, and the corpus of radiocarbon dates produced by archaeological research. However, several biases can affect assemblages of dates, complicating inferences about population size (Figure 1). One serious but addressable issue is landscape taphonomy — the ways in which geologic processes structure the preservation and recovery of archaeological sites and/or materials at landscape scales (Contreras and Codding 2024).

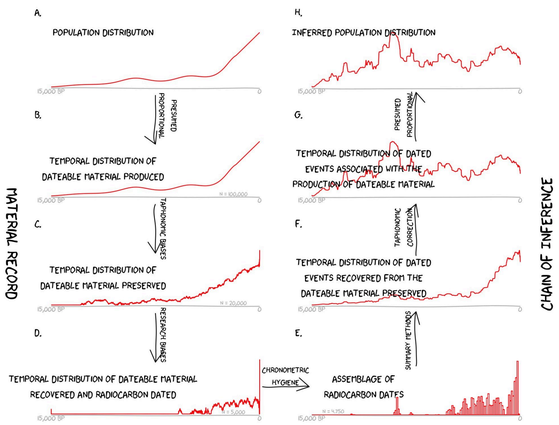
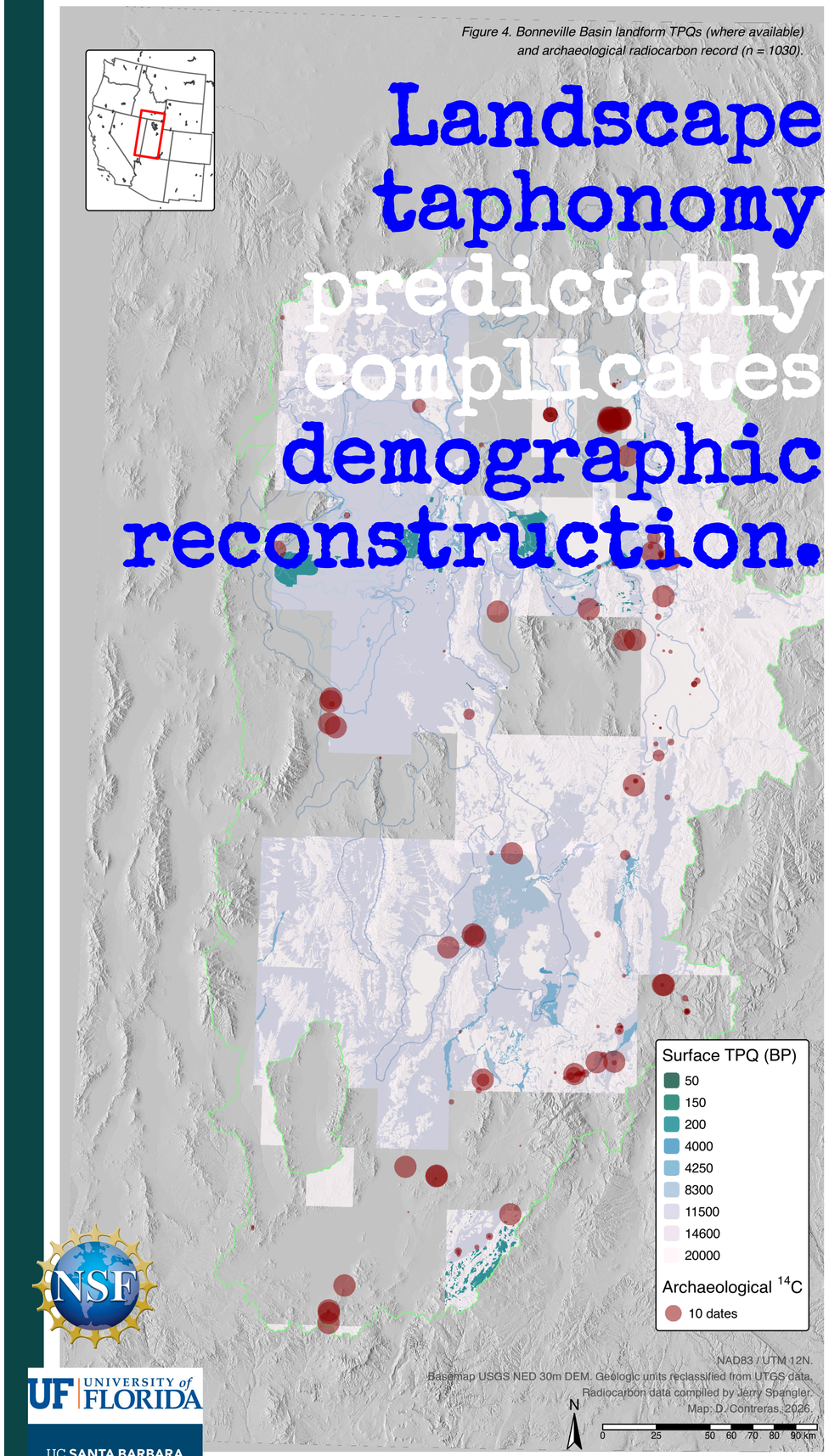


Figure 1: The logic of dates-as-data (Contreras & Codding 2024).

A Bonneville Basin case study

We explore the influence of landscape taphonomy on demographic proxies by using landform ages to assess the proportions of past landscapes that are available to archaeological survey. This landscape taphonomy has structured the radiocarbon record of the Bonneville Basin, and consequently dates-as-data (and other) approaches to estimating Paleoindian populations.



Methods: Landform frequencies

Two primary processes have shaped the Bonneville Basin since the Last Glacial Maximum: a) lake levels, and b) aeolian transport and deposition. Whether archaeological material *was deposited* is structured by (a); whether archaeological material *gets recovered* is (to a large degree) structured by (b).

We explore the effects of these processes by compiling the Utah Geological Survey ages on geologic units in the Bonneville Basin ("Utah Geological Survey Geologic Maps," n.d.), coupled with post-LGM lakestands and Holocene fluctuations summarized by Oviatt and colleagues (2015), to classify the Bonneville Basin landscape by *terminus post quem* (*TPQ*) age (Figure 2).



Figure 2: Project workflow.

The frequency distribution of TPQs provides an empirical correction for frequency-over-time assessments of archaeological material - analogous to Surovell's (Surovell and Brantingham 2007; Surovell et al. 2009; Bluhm and Surovell 2018) proposed global taphonomic correction, but capturing Bonneville Basin landscape taphonomy.

Results: Local taphonomic correction

Taking into account the variation in taphononomic bias over time produces an estimated temporal distribution that is less vulnerable to under-representation, and less prone to over-correction (Figure 3).

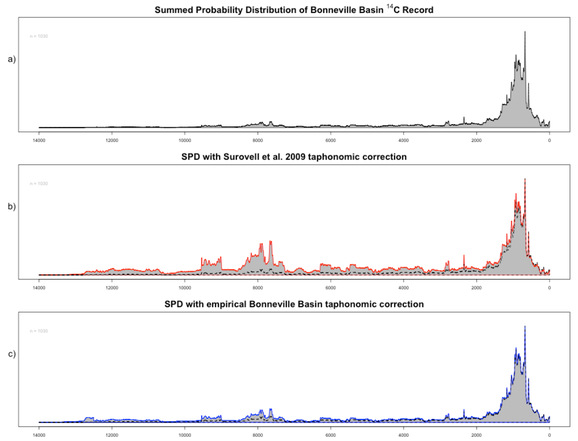


Figure 3: Distributions derived from the Bonneville Basin radiocarbon record (n=1030): a) summed probability distribution, b) summed probability distribution adjusted with the global taphonomic correction from Surovell et al. (2009), and c) summed probability distribution adjusted with an empirically derived Bonneville Basin taphonomic correction. The local taphonomic correction employed in (c) is derived from the frequency distribution of landforms ages in the Bonneville Basin based on published UTGS data, combined with an approximated moderate rate of taphonomic decay following the exponential curve suggested by Surovell & Brantingham (2007).

Even with incomplete coverage, and with much of what is covered classifiable only at coarse resolution, it is clear that the Bonneville Basin archaeological radiocarbon record (Figure 4) is affected by landscape taphonomy. Are we underestimating Paleoindian populations as a result? The radiocarbon record (affected also by research bias) suggests that we are…but not by a lot.

References

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